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End-to-end traceability in Ayush drug supply chains: Challenges, opportunities, and policy perspectives

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ABSTRACT:

Traceability has emerged as a keystone of modern pharmaceutical regulation, ensuring that every product, from raw materials to patient, can be tracked and verified for authenticity, safety, and quality. In India, where the Ayush systems, i.e., Ayurveda, Yoga and Naturopathy, Unani, Siddha, Sowa-Rigpa, and Homeopathy, are expanding rapidly, the need for end-to-end traceability is both a regulatory and public health imperative. It presents a unique traceability challenge for 8369 Ayush manufacturing units (in 2023) due to diverse raw material sources, numerous small and medium-sized producers, and mixed public-private distribution channels. Developed to streamline licensing and regulatory governance, the e-Aushadhi platform could be a key enabler for achieving end-to-end traceability within the Ayush drug supply chain. This review critically examines the current policy and regulatory frameworks to assess the potential for a comprehensive Ayush drug traceability ecosystem. It evaluates the initiatives and measures already implemented, including future opportunities through digital platforms such as e-Aushadhi and e-Charak, and benchmarks them against international traceability systems such as the European Union Falsified Medicines Directive and the US Drug Supply Chain Security Act. The review also explores the transformative potential of emerging technologies, which includes strengthening transparency, authenticity, and accountability across the Ayush drug supply chain. This article suggests an actionable, standards-based, and Artificial Intelligence (AI)-enabled policy roadmap aimed at achieving fully interoperable end-to-end traceability across Ayush drug supply chains.

Keywords:

Artificial Intelligence, Ayush drug supply chain, Blockchain, Drug regulation, Public health policy, Traceability

INTRODUCTION

Traceability in pharmaceutical supply chains refers to the systematic documentation of every step in a product's lifecycle, from raw material sourcing to patient dispensing, enabling verification of authenticity, quality, and safety at all stages.^[1] Globally, robust traceability mechanisms are recognized as fundamental to preventing counterfeit drugs, facilitating recalls, and ensuring regulatory compliance.^[2] For Ayush medicines, traceability presents a unique challenge. The sector's vast diversity in raw

material sourcing, industrial capacity, and distribution networks is the key challenge for consistent quality assurance. Furthermore, issues such as adulteration, misbranding, and spurious drugs have occasionally been reported, undermining consumer trust and international acceptance.^[3] The sector's growing international footprint, supported by rising global demand for Herbal and Traditional Medicines (TMs), further underscores the need for standardized, transparent, and accountable supply chains. Furthermore, sub-standard products have occasionally been reported, which not only compromise patient safety but also impede India's ambition to become a global hub

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for TM. Therefore, ensuring traceability is not merely a regulatory requirement but also a strategic necessity for public health, economic growth, and international acceptance.

Technological innovations are increasingly being recognized as essential enablers of end-to-end traceability. Platforms such as e-Aushadhi, designed for digital licensing and e-Charak, focused on the availability and facilitation of raw material, contributed to improved operational transparency and regulatory compliance. While not primarily traceability systems, these platforms provide valuable data that can support future initiatives for end-to-end traceability. These platforms can be utilized to develop or integrate Artificial Intelligence (AI)-driven analytics and blockchain-enabled record-keeping, providing a foundation to strengthen supply chain monitoring, regulatory oversight, and overall accountability.

CURRENT REGULATORY FRAMEWORK IN INDIA

Regulatory framework overview

The Indian regulatory framework for Ayush drugs is multilayered, combining statutory, administrative, and technical oversight mechanisms. The Drugs and Cosmetics Act, 1940, and associated rules form the backbone of regulation, mandating licensing, quality control, and compliance inspections. In India, Ayush drugs are regulated under the Drugs and Cosmetics Act, 1940, with the Ministry of Ayush (MoA) setting National policies and State Licensing Authorities ensuring regional compliance. At the state level, State Drug Licensing Authorities are responsible for implementing licensing provisions, conducting inspections, and monitoring compliance with pharmacopoeial standards.

Pharmacopoeial standards

In addition, the Pharmacopoeia Commission for Indian Medicine and Homoeopathy (PCIMH) is tasked with developing and updating pharmacopoeial standards for Ayurveda, Siddha, Unani, and Homeopathy (ASU and H), which are mandatory for manufacturers to adhere to and these standards cover raw material quality, formulation procedures, labeling, and packaging.^[4]

Good manufacturing practices

Good Manufacturing Practices (GMP) are legally mandated and form the core quality assurance mechanism for Ayush drugs. Compliance with GMP requires proper documentation of raw material sourcing, standardized formulation processes, quality control testing, and proper storage and distribution practices.

Certificate of Pharmaceutical Product

Certificate of Pharmaceutical Product (CoPP) under the World Health Organization (WHO) GMP Certification Scheme is a crucial regulatory endorsement for the international acceptance of Ayush drugs. Issued by the Central Drugs Standard Control Organization (CDSCO) in India, the CoPP certifies that a particular Ayurveda product adheres to established quality, safety, and efficacy standards as detailed in the Ayurvedic Pharmacopoeia of India (API), and fulfills the requirements for both domestic registration and export to global markets.^[5] The CoPP plays a significant role in facilitating marketing authorizations, ensuring product credibility, and promoting the global competitiveness of Ayush medicines, aligning with WHO benchmarks for herbal product quality. While large-scale manufacturers generally comply with these standards, many small-scale manufacturers face challenges due to limited resources, technical knowledge/expertise, and digital infrastructure. Due to these constraints, they often rely on local suppliers for raw materials, which often supply them with substandard or adulterated products in extreme cases. Out of a total of 8954 manufacturing companies in the year 2019–20, only about 50 companies had revenue above INR 100 crore.^[6] This uneven profit distribution stems from structural gaps: large firms have better capital, modern GMP-compliant facilities, skilled teams, and strong supply chains that help them meet WHO-GMP and CoPP standards and access wider markets. In contrast, most small manufacturers lack adequate resources, technical expertise, and robust quality systems and often depend on inconsistent raw-material suppliers, leading to quality issues and regulatory non-compliance.

Pharmacovigilance

Postmarket safety of Ayush drugs is another important area that cannot be overlooked. The Pharmacovigilance Programme for Ayush Drugs, formally called the Pharmacovigilance Programme for ASU and H Drugs, was launched in 2017, aiming to curb Misleading Advertisements (MLAs) and monitor Adverse Drug Reactions (ADRs). In 2021–22, the initiative was merged into the broader Ayush Oushadhi Gunvatta evam Utpadan Samvardhan Yojana (AOGUSY), a Central Sector Scheme (CSS) covering the years 2021–26, under which pharmacovigilance was made a key component along with the quality improvement of Ayush drugs. The program continues to operate through its three-tier network of one National Pharmacovigilance Coordination Center (NPvCC), five Intermediary Pharmacovigilance Centers (IPvCs), and around 99 Peripheral Pharmacovigilance Centers as of July 2025.^[7] The key regulatory bodies collectively ensure effective governance of the Ayush drug ecosystem through well-defined roles and responsibilities [Figure 1].

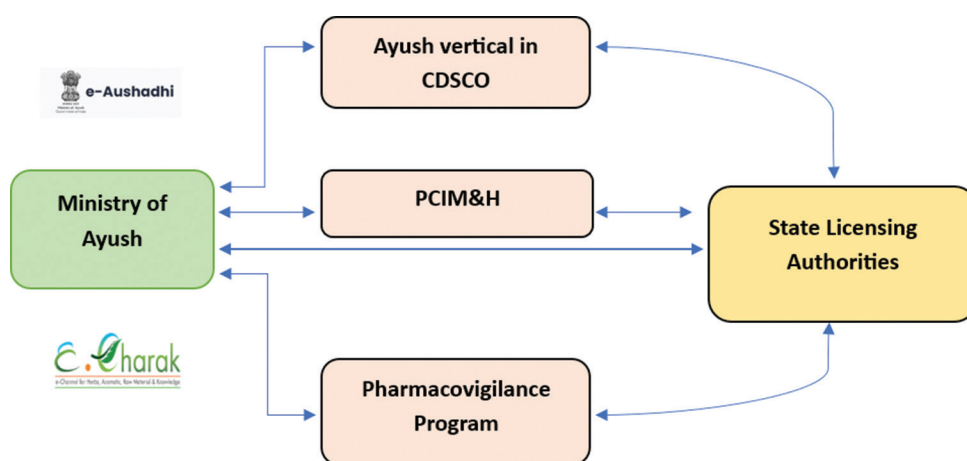


Figure 1: Overview of key bodies regulating the Ayush drug ecosystem. CDSCO = Central Drugs Standard Control Organization, PCIM and H = Pharmacopoeia Commission for Indian Medicine and Homoeopathy

TECHNOLOGICAL INITIATIVES FOR AYUSH DRUG ECOSYSTEM

Technological interventions have become increasingly critical for strengthening Ayush drug supply chains. Among these, e-Aushadhi, an integrated digital platform developed by the MoA, plays a central role by facilitating online licensing and regulatory compliance monitoring.^[8] It ensures transparent tracking of license status and provides real-time information on licensed units, approved formulations, and suspended or cancelled licenses. By enabling end-to-end digital submission, processing, and monitoring of applications, it significantly enhances regulatory efficiency, accountability, and ease of doing business for Ayush drug manufacturers. The system also strengthens oversight by supporting compliance verification, reducing paperwork, and helping curb the circulation of sub-standard products.

The e-Charak portal, developed by the National Medicinal Plants Board, Ministry of Ayush, is a digital platform designed to enable information exchange between various stakeholders involved in medicinal plant sector. It facilitates the exchange of information among farmers, traders, researchers, and entrepreneurs, promoting sustainable cultivation practices and ensuring the quality of raw materials used in Ayush medicines. The platform functions as an online marketplace, allowing buyers and sellers to post and view listings for Medicinal plants and Aromatic herbs, thereby streamlining trade and reducing intermediaries. Detailed product information, including botanical names, sourcing locations, and quality parameters, is submitted and verified through an approval process, contributing to quality assurance. It also provides a repository of agro techniques, best practices, and market price information, supporting stakeholders in adopting scientifically validated and environmentally sustainable methods. By recording

sourcing and quality data digitally, the portal facilitates traceability. The platform improves market access for small-scale producers, supporting economic growth and fair-trade practices. In addition, the data generated through this platform can inform evidence-based policy decisions for regulating and promoting medicinal plant cultivation. It plays a pivotal role in modernizing the Ayush ecosystem, ensuring the authenticity, quality, and sustainability of TM.^[9,10]

The Ayush Suraksha Portal is also a key component of the pharmacovigilance framework for ASU and H medicines. It enables systematic reporting and monitoring of ADRs and MLAs through a decentralized network of peripheral and IPvCs linked to the NPvCC. Integrated into the Ayush Grid, the portal allows seamless data exchange among regulatory authorities, healthcare providers, and consumers, supporting real-time monitoring and informed decision-making. It strengthens the credibility, safety, and accountability of the Ayush sector in India.^[11]

Emerging technologies such as AI and blockchain have the potential to further transform the Ayush supply chain. AI algorithms can analyze production and distribution data to predict risks of contamination, adulteration, or supply disruptions. Predictive models can also optimize inventory management and facilitate targeted inspections, reducing regulatory burden while enhancing compliance.^[12] Blockchain, on the other hand, provides an immutable ledger of transactions across the supply chain. By recording each step from raw material sourcing to dispensing on a distributed ledger, blockchain ensures tamper-proof traceability and strengthens consumer confidence in the authenticity of products. Combining AI and Blockchain with platforms like e-Aushadhi and e-Charak could enable automatic verification of drug quality, real-time detection of anomalies, and enhanced interoperability between manufacturers, regulators, and distributors. Furthermore, the Ayush Suraksha

Portal complements these technologies by providing centralized safety data on ADRs and MLAs, enabling a comprehensive, technology-enabled, and accountable Ayush supply chain.

AYUSH PRODUCTS MARKET AND NEED FOR END-TO-END TRACEABILITY

Ayush sector has demonstrated substantial economic growth in recent years, reflecting a global resurgence of interest in traditional and holistic health systems. According to the Forum on Indian TM under the Research and Information System for Developing Countries, the Indian Ayush industry was valued at USD 18.1 billion in 2020, marking a significant rise from USD 2.85 billion in 2014–15, thereby registering a Compound Annual Growth Rate of approximately 17% during 2014–2020.^[13] This is further valued at USD 43.4 billion in 2023. Similarly, Ayush exports have doubled from USD 1.09 billion (2014) to USD 2.16 billion (2023).^[14]

This rapid expansion underscores both the economic potential and regulatory responsibility associated with the Ayush sector. As the market continues to diversify, encompassing classical formulations, proprietary medicines, dietary supplements, and herbal cosmetics, ensuring product quality, authenticity, and consumer safety becomes paramount. The heterogeneous nature of the Ayush supply chain, spanning raw material collection, processing, manufacturing, packaging, distribution, and retail, makes it particularly susceptible to quality inconsistencies, sub-standard products, and documentation gaps. These challenges underline the necessity for end-to-end traceability systems, capable of tracking each product's journey from source to end user. A robust traceability mechanism would not only strengthen regulatory oversight and consumer confidence but also facilitate India's alignment with international quality standards, as advocated by the WHO Global Strategy for TM (2025–2034).

INTERNATIONAL PRACTICES FOR TRACEABILITY

End-to-end traceability in TM is increasingly recognized as a critical mechanism to ensure product authenticity, safety, and efficacy throughout the supply chain, from cultivation and processing to distribution and consumption. International experiences provide valuable insights for developing robust systems across diverse TM sectors. In China, traceability of Traditional Chinese Medicine (TCM) has been enhanced through blockchain-based platforms that record the entire lifecycle of medicinal products, including cultivation, processing, storage, and distribution. This system creates an immutable ledger, ensuring data integrity

and transparency, while preventing the circulation of counterfeit or substandard products.^[15] Studies have shown that such blockchain-integrated systems improve consumer trust, facilitate regulatory compliance, and enhance the overall quality of TMs.

In the European Union, the Falsified Medicines Directive (FMD, 2011/62/EU) mandates serialization and tamper-evident packaging of prescription medicines, including herbal products.^[16] The European Medicines Verification System enables verification at multiple points in the supply chain, allowing rapid identification and removal of suspect or falsified medicines (European Medicines Agency, 2019). Similarly, in the United States, the Drug Supply Chain Security Act (DSCSA, 2013) requires unique product identifiers, electronic verification, and transaction reporting, emphasizing interoperability across manufacturers, wholesalers, and dispensers to maintain supply chain integrity.^[17] These frameworks illustrate how regulatory mandates, when coupled with digital infrastructure, can ensure robust traceability while supporting global trade and public safety.

Other international initiatives further demonstrate innovative approaches to traceability. In South Korea, QR code-based verification systems enable consumers and healthcare providers to access detailed information about the origin, composition, and manufacturing processes of herbal medicines.^[18] Canada has piloted blockchain-enabled traceability for herbal products, recording cultivation, processing, and distribution data to reduce counterfeiting risks. Collectively, these initiatives highlight that technologies such as blockchain, QR codes, and electronic verification, when integrated with standardized regulatory frameworks, are highly effective in ensuring the safety, authenticity, and quality of TMs on a global scale.

The WHO has emphasized the importance of traceability in TM, recommending strategies such as serialization, electronic reporting, and interoperability among stakeholders to enhance product safety and efficacy. Scientific studies also indicate that effective traceability systems rely on key elements, including standardized data collection, real-time monitoring, and collaboration among regulators, manufacturers, and distributors.^[15] By adopting international best practices and leveraging advanced technologies, the global TM sector can significantly reduce the prevalence of substandard products, enhance consumer confidence, and facilitate international acceptance. End-to-end traceability thus emerges as a critical pillar for ensuring the sustainability, reliability, and globalization of TMs.^[19] Despite these advancements, some gaps remain, including low digital adoption among small manufacturers, limited

digital literacy, fragmented data systems, inconsistent raw-material traceability, and uneven integration of state regulatory mechanisms.

CHALLENGES AND OPPORTUNITIES

Ayush sector has made remarkable strides in regulatory frameworks, quality assurance, and digitalization, reflecting India's commitment to safe, effective, and globally competitive TMs. The development of platforms for monitoring, licensing, and quality control demonstrates a strong foundation for a contemporary, transparent supply chain. These achievements provide a promising starting point to further strengthen end-to-end traceability, quality assurance, and consumer trust.

Despite this progress, several challenges persist. A primary concern lies in the fragmented nature of raw material sourcing. Most medicinal plants are wild harvested or cultivated on a small scale, leading to variability in botanical identification and sustainability practices. Without standardized procurement protocols and robust digital verification, traceability from source to manufacturer remains limited, increasing the risk of sub-standard products and quality inconsistencies. However, these challenges also create significant opportunities for innovation and modernization. Adoption of AI and predictive analytics can support verification of raw materials, early detection of quality deviations, and mapping of supply trends for better planning. Blockchain technology can provide tamper-proof, immutable records, enabling end-to-end traceability and transparency. The potential challenges and corresponding opportunities can be outlined as follows:

Challenges

Fragmentation of the industry

A substantial proportion of Ayush drug manufacturers operate on a small or micro scale, often with limited access to digital infrastructure and regulatory guidance. This fragmentation leads to inconsistent data capture, recordkeeping, and reporting, making it difficult to monitor compliance uniformly. The lack of standardization across manufacturing units hinders the creation of a cohesive, auditable supply chain. As per the "All India Trade Survey of Prioritized Medicinal Plants (2019)," there is increased demand for high-value medicinal plants. This demand has increased by 50% and the availability of these plants has decreased by 26%. The low participation of Indian herbals can be attributed to low-quality processes for agriculture and quality control procedures, a lack of modern methods for processing, limited R and D, inadequate mechanisms for product and process standardization, and weak regulatory frameworks involved in medicinal plant trade.^[20]

Complex and heterogeneous supply chains

Ayush drugs are distributed through a mix of public and private channels, alongside informal markets and local vendors. This complexity creates vulnerabilities, including sub-standard and diversion of products, which compromise quality and consumer trust. In addition, tracing products back to raw material sources is challenging, making recall management and quality assurance more difficult.

Limited technical expertise and digital literacy

Many stakeholders, particularly in small-scale manufacturing units and rural procurement regions, lack the skills and training required to utilize advanced digital tools such as AI, Blockchain, or Internet of Things (IoT)-enabled monitoring systems. This limits the effective implementation of technology-driven solutions for compliance, quality assurance, and real-time oversight.

Regulatory gaps and implementation challenges

While platforms like e-Aushadhi and e-Charak exist, regulatory oversight is uneven due to inconsistent enforcement, limited integration across state and central bodies. These gaps reduce the effectiveness of existing traceability, quality, and pharmacovigilance frameworks.

Unorganized production and cultivation of medicinal plants in India

There is an unequipped supply chain and its management in the Indian ecosystem. There is a need to build an adequately managed and regulated supply chain for the production, cultivation, and exploration of medicinal plants in the country. There is also a need to build farmer associations working with medicinal plants.

Lack of entrepreneurship and innovation in the field of the traditional medicine system

There is a need for entrepreneurship and innovation to speed up awareness and exploration of the TM system. The advent of startups in this field will enhance the TM system in India. The startups about medicinal plants can act as a route to bring adequate farming techniques along with integrating modern applications such as AI and data analytics with the crop profiling, crop and seed analysis, harvest, storage, and trading.

Opportunities

Technological modernization

Integrating AI, blockchain, and IoT into Ayush supply chains can transform monitoring and compliance mechanisms. AI can support predictive analysis for quality deviations, contamination and automate inspections. Blockchain ensures tamper-proof and transparent recordkeeping, and IoT devices enable real-time tracking of inventory, transport, and storage

conditions. Together, these technologies can strengthen supply chain integrity.

Capacity-building and skill development

Focused training programs for small and medium-scale manufacturers, distributors, and regulatory personnel can enhance digital literacy and technological adoption. Empowering stakeholders to use digital platforms efficiently ensures broader compliance and promotes a culture of quality and accountability across the sector.

Platform integration and international alignment

e-Aushadhi and e-Charak provide a strong foundation for digital monitoring, which can be further enhanced by linking with international standard-compliant frameworks such as the EU FMD and the US DSCSA. This enables global benchmarking, harmonized reporting, and improved credibility of Ayush products in international markets.

Standardization across the supply chain

Establishing uniform standards for raw material sourcing, formulation practices, labeling, and quality testing can reduce variability and strengthen consumer confidence. Standardization supports regulatory oversight, risk mitigation, and traceable accountability, while also facilitating export readiness and wider market acceptance.

Data-driven policy and decision making

Digital platforms can generate real-time analytics on production, distribution, and adverse events, enabling policymakers to design targeted interventions, optimize resource allocation, and respond promptly to quality or safety concerns. This creates a more resilient and proactive regulatory ecosystem.

POLICY RECOMMENDATIONS AND ROADMAP

To modernize the Ayush pharmaceutical ecosystem and ensure global competitiveness, there is a pressing need to establish a transparent, digitally enabled, and interoperable supply chain framework. The following recommendations outline a comprehensive approach to achieving the same:

Policy and regulatory actions

- I. Strengthen harmonization between central and state regulatory authorities to ensure uniform implementation of traceability policies and prevent regulatory gaps
- II. Institute digital tracking of Ayush drug batches from raw material sourcing to patient dispensing to achieve interoperable, AI-enabled end-to-end traceability and integrate platforms such as e-Aushadhi and e-Charak

with AI-driven analytics and blockchain modules, making supply chain anomalies detectable and end-to-end traceability transparent, even for small and medium producers

- III. Align traceability standards with international frameworks by incorporating serialization, tamper-evident packaging, and electronic verification mechanisms
- IV. Develop and implement a robust national roadmap to establish a modern Ayush supply chain ecosystem that ensures safety, authenticity, and quality of TMs while enhancing international market access and consumer confidence.

Infrastructure and digital capacity building

- I. Implement capacity-building programs for small and medium manufacturers to ensure GMP compliance, digital literacy, and data reporting capabilities
- II. Link manufacturers, distributors, and retailers through standardized reporting protocols integrated with e-Aushadhi and e-Charak platforms
- III. Encourage public-private partnerships to mobilize investments in digital infrastructure for the Ayush drug supply chain
- IV. Invest in rural and peri-urban digital connectivity to bridge supply chain gaps.

Technology adoption and integration

- I. Incentivize the Ayush drug industry for the adoption of emerging technologies such as AI, blockchain, and IoT – under schemes like AOGUSY to enable predictive monitoring, fraud detection, and tamper-proof verification
- II. Integrate e-Aushadhi and e-Charak, and Pharmacovigilance portals for unified transaction data.

Monitoring, evaluation, and research

- I. Institutionalize third-party, multistakeholder compliance audits
- II. Promote research and development initiatives for digital traceability, development of interoperable data standards, and integration of pharmacovigilance systems to enhance safety monitoring
- III. Foster collaboration with international agencies to align India's traceability framework with global quality and regulatory standards.

Alignment with global policy initiatives

The integration of TM into modern healthcare systems is also in alignment with global health policy initiatives.^[21] The sociocultural and biodiversity heritages of TM are invaluable resources to evolve inclusive, diverse, and equitable approaches to sustainable development.

Expanding from local practices to global phenomena, TM is part of the growing trillion-dollar health, wellness, beauty, and pharmaceutical industries.^[22] Over 40% of conventional pharmaceutical formulations are based on natural products.^[23] Current trends toward more holistic, personalized, and preventative medicine would be well supported by policies in TMs globally, as the “Astana Declaration” adopted at the Global Conference on Primary Health Care (PHC) in 2018, reinforces the role of PHC in achieving Universal Health Coverage and the Sustainable Development Goals. It emphasizes the integration of TMs, improved accessibility, patient safety, and the use of digital technologies to enhance health service delivery and data transparency.^[24]

The “2030 Agenda for Sustainable Development” also highlights the transformative potential of Information and Communication Technologies in reducing the digital divide and empowering individuals to engage in their healthcare through digital tools.^[25] This agenda also discusses integrating digital technologies into TM to provide greater access to traditional practices and promoting a more inclusive, citizen-centric healthcare model. The WHO has significantly advanced integration of TM with digital technologies through standardization and digitalization efforts. The key elements and technologies that enable complete traceability in the Ayush drug supply chain are illustrated in Figure 2.

CONCLUSION

End-to-end traceability is vital to ensure the authenticity, safety, and quality of Ayush drugs, strengthen consumer confidence, and enhance global market access. India has made notable progress through regulatory reforms, pharmacopoeial standardization, and digital initiatives such as e-Aushadhi and e-Charak, which support licensing and availability of raw material. However, persistent challenges remain, including fragmented raw material supply chains, limited interoperability between central and state systems, and inconsistent data capture. These gaps restrict real-time monitoring and full transparency.

Leveraging AI for predictive quality analytics and blockchain for tamper-proof documentation can transform traceability into a scalable, reliable, and transparent system. Integrating these technologies within existing regulatory frameworks, coupled with strengthened inter-agency coordination and capacity-building, will advance a unified and interoperable digital ecosystem. A robust, technology-driven traceability framework will not only safeguard public health and curb adulteration but also elevate India’s leadership in the global TM market, promoting sustainability, credibility, and international competitiveness of the Ayush sector.

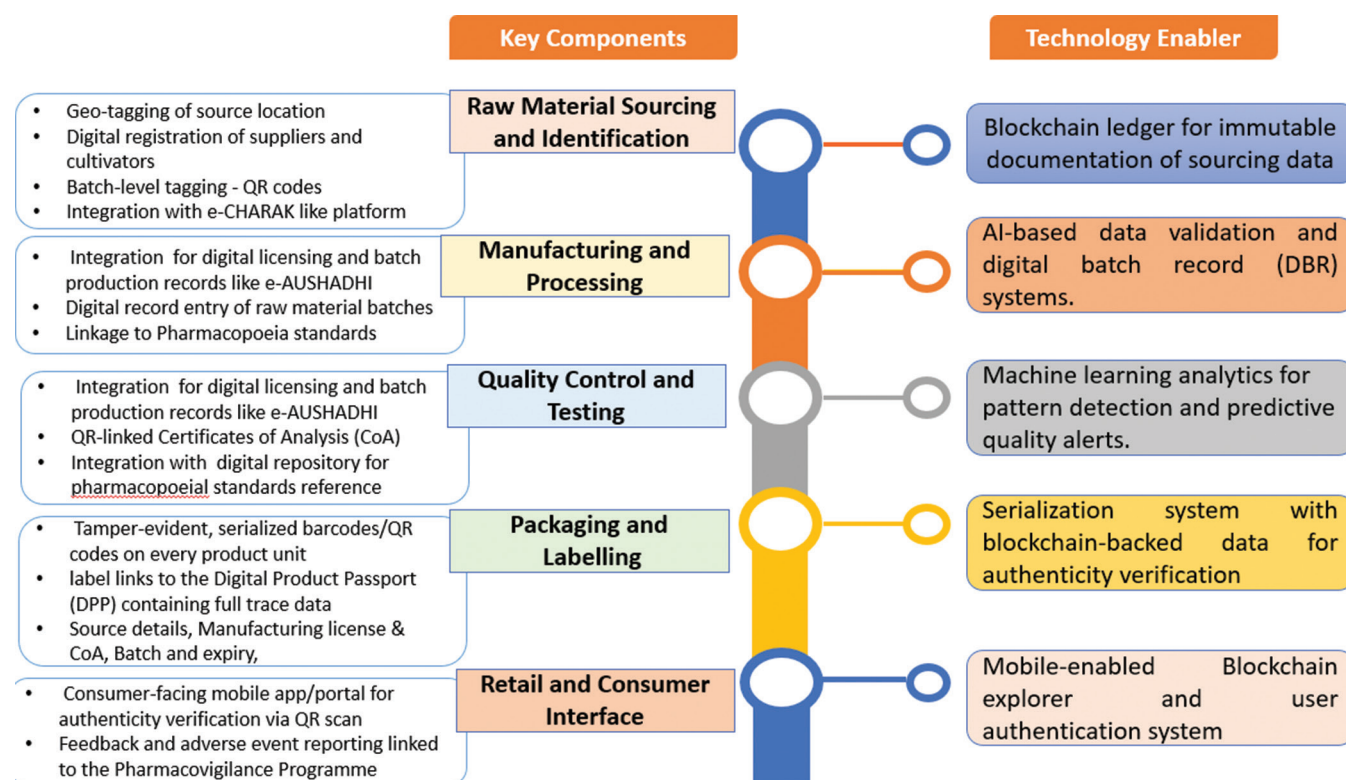


Figure 2: Framework for end-to-end traceability in Ayush drug supply chains: Key components and technology enablers

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Conflicts of interest

There are no conflicts of interest.

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